

Random Walks and Electrical Networks: Harmonic Functions, Voltage, Effective Resistance, and Rayleigh–Thomson Theory

1 Weighted graphs and random walks

1.1 Networks

Definition 1.1 (Electrical network). *An electrical network is a connected, locally finite graph*

$$G = (V, E)$$

together with a positive conductance

$$c_{xy} = c_{yx} > 0$$

for every edge $xy \in E$.

The corresponding resistance of the edge xy is

$$r_{xy} = \frac{1}{c_{xy}}.$$

Locally finite means that each vertex has finitely many neighbors. For a vertex x , define its total conductance by

$$c_x = \sum_{y \sim x} c_{xy}.$$

The graph G together with c_{xy} determines a random walk by

$$p(x, y) = \frac{c_{xy}}{c_x}, \quad y \sim x.$$

Thus the random walk jumps from x to y with probability proportional to the conductance of the edge xy .

Example 1.2 (Simple random walk). *If every edge has conductance $c_{xy} = 1$, then*

$$c_x = \deg(x),$$

and hence

$$p(x, y) = \frac{1}{\deg(x)}.$$

This is the usual simple random walk on a graph.

1.2 Reversibility

The random walk defined above is reversible with respect to the measure c_x .

Indeed,

$$c_x p(x, y) = c_x \frac{c_{xy}}{c_x} = c_{xy} = c_{yx} = c_y p(y, x).$$

So

$$\boxed{c_x p(x, y) = c_y p(y, x).}$$

This reversibility is one reason electrical networks are naturally connected to random walks.

2 Harmonic functions

2.1 Definition

Definition 2.1 (Harmonic function). *Let $G = (V, E)$ be a weighted graph. A function $h : V \rightarrow \mathbb{R}$ is harmonic at a vertex x if*

$$h(x) = \sum_{y \sim x} p(x, y) h(y).$$

Equivalently,

$$h(x) = \frac{1}{c_x} \sum_{y \sim x} c_{xy} h(y).$$

Equivalently,

$$\sum_{y \sim x} c_{xy} (h(y) - h(x)) = 0.$$

The last expression motivates the weighted graph Laplacian.

Definition 2.2 (Weighted graph Laplacian). *For a function $f : V \rightarrow \mathbb{R}$, define*

$$\Delta f(x) = \sum_{y \sim x} c_{xy} (f(y) - f(x)).$$

Then f is harmonic at x exactly when

$$\Delta f(x) = 0.$$

Remark 2.3. *When G is d -regular and all $c_{xy} = 1$, the harmonic condition becomes*

$$h(x) = \frac{1}{d} \sum_{y \sim x} h(y).$$

This is the familiar statement that $h(x)$ equals the average of its neighbors. For general weighted graphs, however, it is a weighted average.

3 Maximum principle and uniqueness

Let $D \subset V$ be a finite connected set. The boundary of D is

$$\partial D = \{y \notin D : \exists x \in D \text{ with } x \sim y\}.$$

We usually study functions on $D \cup \partial D$ that are harmonic on D .

Theorem 3.1 (Maximum principle). *Let $D \subset V$ be finite and connected. Suppose h is harmonic on D . Then*

$$\max_{D \cup \partial D} h$$

is achieved on the boundary ∂D , unless h is constant on $D \cup \partial D$ in the connected component under consideration.

Similarly, the minimum is achieved on ∂D , unless h is constant.

证明. Let

$$M = \max_{D \cup \partial D} h.$$

Suppose the maximum is achieved at some interior point $x \in D$, so $h(x) = M$.

Since h is harmonic at x ,

$$h(x) = \sum_{y \sim x} p(x, y) h(y).$$

Every term satisfies $h(y) \leq M = h(x)$. Since $h(x)$ is a weighted average of the neighboring values, and all weights $p(x, y)$ are positive and sum to 1, the only way the average can equal the maximum is if

$$h(y) = M$$

for every neighbor $y \sim x$.

Now apply the same argument to each such neighbor that lies in D . By connectedness, this propagates through the whole component. Thus if an interior maximum occurs, the function must be constant throughout the connected region.

The proof for the minimum is identical, or follows by applying the maximum principle to $-h$. □

Theorem 3.2 (Uniqueness for the Dirichlet problem). *Let $D \subset V$ be finite. Suppose h_1, h_2 are harmonic on D , and suppose*

$$h_1 = h_2 \quad \text{on } \partial D.$$

Then

$$h_1 = h_2 \quad \text{on } D \cup \partial D.$$

证明. Let

$$u = h_1 - h_2.$$

Then u is harmonic on D , because the Laplacian is linear:

$$\Delta u = \Delta h_1 - \Delta h_2 = 0.$$

Also,

$$u = 0 \quad \text{on } \partial D.$$

By the maximum principle,

$$\max_{D \cup \partial D} u$$

is achieved on ∂D , unless u is constant. Since $u = 0$ on ∂D , we get

$$u \leq 0.$$

Applying the same argument to $-u$, we get

$$u \geq 0.$$

Therefore $u \equiv 0$, so $h_1 = h_2$. □

4 Hitting probabilities are harmonic

For a set $A \subset V$, define the hitting time

$$T_A = \inf\{n \geq 0 : X_n \in A\}.$$

For a single vertex a , write

$$T_a = T_{\{a\}}.$$

Also define the first return time to a by

$$T_a^+ = \inf\{n \geq 1 : X_n = a\}.$$

Proposition 4.1 (Hitting probabilities are harmonic). *Let $a, b \in V$, $a \neq b$. For $x \in V$, define*

$$h(x) = \mathbb{P}_x(T_a < T_b).$$

Then

$$h(a) = 1, \quad h(b) = 0,$$

and h is harmonic at every $x \neq a, b$.

证明. The boundary values are immediate:

$$h(a) = \mathbb{P}_a(T_a < T_b) = 1,$$

and

$$h(b) = \mathbb{P}_b(T_a < T_b) = 0.$$

Now let $x \neq a, b$. By conditioning on the first step,

$$h(x) = \sum_{y \sim x} \mathbb{P}_x(X_1 = y) \mathbb{P}_y(T_a < T_b).$$

Since $\mathbb{P}_x(X_1 = y) = p(x, y)$, we get

$$h(x) = \sum_{y \sim x} p(x, y) h(y).$$

Thus h is harmonic at x . □

This proposition is the first half of the random walk–electrical network dictionary.

5 Voltage and current

5.1 Voltage

In an electrical network, a voltage is simply a function

$$v : V \rightarrow \mathbb{R}.$$

If we prescribe voltage values at certain boundary vertices and require the function to be harmonic in the interior, then v is determined by the Dirichlet problem.

For example, in a finite network with two distinguished vertices a, b , the voltage v with boundary conditions

$$v(a) = 1, \quad v(b) = 0$$

and harmonicity

$$\Delta v(x) = 0, \quad x \neq a, b,$$

is exactly the hitting probability

$$v(x) = \mathbb{P}_x(T_a < T_b).$$

Theorem 5.1 (Voltage equals hitting probability). *Let G be finite, and fix distinct vertices $a, b \in V$. Let v be the unique function satisfying*

$$v(a) = 1, \quad v(b) = 0,$$

and

$$\Delta v(x) = 0, \quad x \neq a, b.$$

Then

$$\boxed{v(x) = \mathbb{P}_x(T_a < T_b)}.$$

证明. Define

$$h(x) = \mathbb{P}_x(T_a < T_b).$$

By the previous proposition, $h(a) = 1$, $h(b) = 0$, and h is harmonic on $V \setminus \{a, b\}$.

Thus h and v solve the same Dirichlet problem. By uniqueness,

$$h = v.$$

□

This theorem is the fundamental reason voltage is probabilistic.

5.2 Current and Ohm's law

Given a voltage v , define the current along an oriented edge $x \rightarrow y$ by

$$i(x, y) = c_{xy}(v(x) - v(y)).$$

Equivalently,

$$v(x) - v(y) = r_{xy}i(x, y).$$

This is Ohm's law.

Notice that

$$i(y, x) = c_{yx}(v(y) - v(x)) = -i(x, y).$$

So current is antisymmetric on oriented edges.

Definition 5.2 (Antisymmetric edge function). *An edge function θ is antisymmetric if*

$$\theta(x, y) = -\theta(y, x)$$

for every edge $xy \in E$.

Current is an antisymmetric edge function.

5.3 Divergence

Definition 5.3 (Divergence of a flow). *For an antisymmetric edge function θ , define its divergence at x by*

$$\operatorname{div} \theta(x) = \sum_{y \sim x} \theta(x, y).$$

This is the net amount of flow leaving x .

If i comes from a voltage v , then

$$\operatorname{div} i(x) = \sum_{y \sim x} c_{xy}(v(x) - v(y)) = -\Delta v(x).$$

Therefore, v is harmonic at x if and only if

$$\operatorname{div} i(x) = 0.$$

Thus:

harmonic voltage $\iff$ current is conserved at that vertex.
--

This is Kirchhoff's current law.

6 Escape probability and effective resistance

We now connect effective resistance with escape probabilities.

Let G be a finite network, and fix two distinct vertices $a, b \in V$. Think of a as the starting point and b as the target boundary.

Let v be the voltage function satisfying

$$v(a) = 1, \quad v(b) = 0,$$

and

$$\Delta v(x) = 0, \quad x \neq a, b.$$

By the previous theorem,

$$v(x) = \mathbb{P}_x(T_a < T_b).$$

In particular, for a neighbor $y \sim a$,

$$v(y) = \mathbb{P}_y(T_a < T_b).$$

Now define the escape probability from a to b before returning to a by

$$p_{\text{esc}} = \mathbb{P}_a(T_b < T_a^+).$$

Starting from a , the walk first moves to a neighbor y with probability

$$p(a, y) = \frac{c_{ay}}{c_a}.$$

After moving to y , the probability of returning to a before hitting b is

$$\mathbb{P}_y(T_a < T_b) = v(y).$$

Therefore the probability of returning to a before hitting b is

$$\sum_{y \sim a} p(a, y) v(y).$$

Hence the probability of escaping to b before returning to a is

$$p_{\text{esc}} = 1 - \sum_{y \sim a} p(a, y) v(y).$$

Using

$$p(a, y) = \frac{c_{ay}}{c_a},$$

we get

$$p_{\text{esc}} = 1 - \sum_{y \sim a} \frac{c_{ay}}{c_a} v(y).$$

Now compute the total current flowing out of a . Since $v(a) = 1$,

$$i_a = \sum_{y \sim a} c_{ay} (v(a) - v(y)) = \sum_{y \sim a} c_{ay} (1 - v(y)).$$

Equivalently,

$$\begin{aligned} i_a &= \sum_{y \sim a} (v(a) - v(y)) c_{ay} \\ &= \sum_{y \sim a} (v(a) - v(y)) \frac{c_{ay}}{c_a} c_a \\ &= c_a \left(1 - \sum_{y \sim a} p(a, y) v(y) \right) \\ &= c_a p_{\text{esc}}. \end{aligned}$$

Thus we have the fundamental identity

$$\boxed{i_a = c_a p_{\text{esc}}}.$$

Since the voltage difference between a and b is normalized to be 1, the total current i_a is the effective conductance:

$$i_a = C_{\text{eff}}(a, b).$$

Therefore

$$C_{\text{eff}}(a, b) = c_a \mathbb{P}_a(T_b < T_a^+).$$

Equivalently, since

$$R_{\text{eff}}(a, b) = \frac{1}{C_{\text{eff}}(a, b)},$$

we get

$$\mathbb{P}_a(T_b < T_a^+) = \frac{1}{c_a R_{\text{eff}}(a, b)}.$$

This is one of the central formulas connecting random walks and electrical networks.

Remark 6.1. *The factor c_a is important. The formula is not*

$$\mathbb{P}_a(T_b < T_a^+) = \frac{1}{R_{\text{eff}}(a, b)}.$$

The correct formula is

$$\mathbb{P}_a(T_b < T_a^+) = \frac{1}{c_a R_{\text{eff}}(a, b)}.$$

For simple random walk on $\mathbb{Z}^d$, every vertex has total conductance

$$c_a = 2d.$$

Thus, on $\mathbb{Z}^d$,

$$\mathbb{P}_a(T_b < T_a^+) = \frac{1}{2d R_{\text{eff}}(a, b)}.$$

7 Energy of a flow

We now introduce the variational principle behind effective resistance.

7.1 Energy

Let θ be an antisymmetric edge function. Its energy is

$$\mathcal{E}(\theta) = \frac{1}{2} \sum_{x,y} r_{xy} \theta(x, y)^2.$$

Equivalently, if we sum over unoriented edges e , then

$$\mathcal{E}(\theta) = \sum_{e \in E} r_e \theta(e)^2.$$

The two conventions are the same; the factor $1/2$ compensates for counting each edge twice.

7.2 Unit flows

Definition 7.1 (Unit flow from a to b). *A unit flow from a to b is an antisymmetric edge function θ such that*

$$\operatorname{div} \theta(a) = 1,$$

$$\operatorname{div} \theta(b) = -1,$$

and

$$\operatorname{div} \theta(x) = 0 \quad \text{for all } x \neq a, b.$$

So θ sends one unit of current out of a , absorbs one unit at b , and conserves current everywhere else.

8 Conservation of energy and cycle law

Suppose a current i comes from a voltage v , so

$$i(x, y) = c_{xy}(v(x) - v(y)).$$

Equivalently,

$$r_{xy}i(x, y) = v(x) - v(y).$$

Let

$$x_0, x_1, \dots, x_n = x_0$$

be a cycle in the graph. Then

$$\sum_{k=0}^{n-1} (v(x_k) - v(x_{k+1})) = 0$$

because the sum telescopes.

Using Ohm's law, this becomes

$$\boxed{\sum_{k=0}^{n-1} r_{x_k x_{k+1}} i(x_k, x_{k+1}) = 0.}$$

This is Kirchhoff's voltage law, also called the cycle law.

The interpretation is:

Around a closed loop, the total voltage drop is zero.

This is sometimes described as conservation of energy: going around a closed loop cannot create or destroy net potential energy.

9 Energy identity for electrical current

Let i be the electrical current induced by a voltage v . Then

$$i(x, y) = c_{xy}(v(x) - v(y)).$$

Thus

$$r_{xy}i(x, y)^2 = i(x, y)(v(x) - v(y)).$$

Summing over unoriented edges,

$$\mathcal{E}(i) = \sum_{e=(x,y)} i(e)(v(x) - v(y)).$$

By discrete integration by parts,

$$\mathcal{E}(i) = \sum_{x \in V} v(x) \operatorname{div} i(x).$$

Now suppose i is the electrical current from a to b , and the total current is I . Then

$$\operatorname{div} i(a) = I,$$

$$\operatorname{div} i(b) = -I,$$

and

$$\operatorname{div} i(x) = 0 \quad \text{for } x \neq a, b.$$

Therefore,

$$\mathcal{E}(i) = v(a)I + v(b)(-I) = I(v(a) - v(b)).$$

So we get the energy identity

$$\mathcal{E}(i) = I(v(a) - v(b)).$$

This is the electrical power identity $P = IV$.

9.1 Two normalizations

There are two useful ways to normalize.

Voltage difference 1

If

$$v(a) - v(b) = 1,$$

then

$$I = C_{\text{eff}}(a, b).$$

The energy identity becomes

$$\mathcal{E}(i) = C_{\text{eff}}(a, b).$$

Unit current

If

$$I = 1,$$

then

$$v(a) - v(b) = R_{\text{eff}}(a, b).$$

The energy identity becomes

$$\boxed{\mathcal{E}(i) = R_{\text{eff}}(a, b).}$$

This is the form used in Thomson's principle.

10 Thomson's principle

Thomson's principle says that the actual electrical current is the most energy-efficient way to send one unit of current from a to b .

Theorem 10.1 (Thomson's principle). *Let G be a finite electrical network, and let $a, b \in V$. Then*

$$\boxed{R_{\text{eff}}(a, b) = \inf_{\theta} \mathcal{E}(\theta),}$$

where the infimum is taken over all unit flows θ from a to b .

Moreover, the minimizer is the physical electrical unit current.

证明. Let i be the physical electrical current from a to b , normalized so that it is a unit current:

$$\operatorname{div} i(a) = 1, \quad \operatorname{div} i(b) = -1, \quad \operatorname{div} i(x) = 0 \quad (x \neq a, b).$$

By the energy identity,

$$\mathcal{E}(i) = R_{\text{eff}}(a, b).$$

Now let θ be any other unit flow from a to b . Define

$$\eta = \theta - i.$$

Since both θ and i have the same divergence, we have

$$\operatorname{div} \eta(x) = 0 \quad \text{for every } x \in V.$$

Thus η is a circulation: it has no source and no sink.

Now expand the energy:

$$\mathcal{E}(\theta) = \mathcal{E}(i + \eta) = \sum_e r_e (i(e) + \eta(e))^2.$$

Expanding the square,

$$\mathcal{E}(\theta) = \sum_e r_e i(e)^2 + 2 \sum_e r_e i(e) \eta(e) + \sum_e r_e \eta(e)^2.$$

Thus

$$\mathcal{E}(\theta) = \mathcal{E}(i) + 2 \sum_e r_e i(e) \eta(e) + \mathcal{E}(\eta).$$

We claim the middle term is zero.

Since i is an electrical current, it comes from a voltage v . Therefore

$$r_e i(e) = v(x) - v(y)$$

for $e = (x, y)$. Hence

$$\sum_e r_e i(e) \eta(e) = \sum_{e=(x,y)} (v(x) - v(y)) \eta(e).$$

By discrete integration by parts,

$$\sum_{e=(x,y)} (v(x) - v(y)) \eta(e) = \sum_{x \in V} v(x) \operatorname{div} \eta(x).$$

But

$$\operatorname{div} \eta(x) = 0 \quad \text{for all } x.$$

Therefore,

$$\sum_e r_e i(e) \eta(e) = 0.$$

So

$$\mathcal{E}(\theta) = \mathcal{E}(i) + \mathcal{E}(\eta).$$

Since

$$\mathcal{E}(\eta) \geq 0,$$

we get

$$\mathcal{E}(\theta) \geq \mathcal{E}(i).$$

Thus i minimizes energy among all unit flows.

Finally,

$$\mathcal{E}(i) = R_{\text{eff}}(a, b).$$

Therefore,

$$R_{\text{eff}}(a, b) = \inf_{\theta} \mathcal{E}(\theta).$$

□

Remark 10.2 (Physical meaning). *Any unit flow θ can be decomposed as*

$$\theta = i + \eta,$$

where i is the real electrical current and η is a useless circulation. The circulation does not help transport current from a to b , but it dissipates extra energy. Therefore the true electrical current is the energy minimizer.

11 Rayleigh monotonicity law

Rayleigh monotonicity says that increasing resistances cannot decrease effective resistance.

Theorem 11.1 (Rayleigh monotonicity). *Let G be a finite network with edge resistances r_e . Let G' be another network on the same graph with resistances r'_e . If*

$$r'_e \geq r_e \quad \text{for every edge } e,$$

then

$$\boxed{R_{\text{eff}G'}(a, b) \geq R_{\text{eff}G}(a, b).}$$

证明. By Thomson's principle,

$$R_{\text{eff}G}(a, b) = \inf_{\theta} \sum_e r_e \theta(e)^2,$$

where the infimum is over all unit flows from a to b .

Similarly,

$$R_{\text{eff}G'}(a, b) = \inf_{\theta} \sum_e r'_e \theta(e)^2.$$

For every fixed unit flow θ , since $r'_e \geq r_e$, we have

$$\sum_e r'_e \theta(e)^2 \geq \sum_e r_e \theta(e)^2.$$

Taking infimum over all unit flows gives

$$\inf_{\theta} \sum_e r'_e \theta(e)^2 \geq \inf_{\theta} \sum_e r_e \theta(e)^2.$$

Therefore,

$$R_{\text{eff}G'}(a, b) \geq R_{\text{eff}G}(a, b).$$

□

11.1 Cutting

Cutting means deleting edges. Deleting an edge can be viewed as increasing its resistance to infinity:

$$r_e \mapsto \infty.$$

Therefore, by Rayleigh monotonicity,

$$\boxed{\text{cutting edges can only increase effective resistance.}}$$

Thus if $H \subset G$ is a subgraph obtained by deleting edges, then

$$\boxed{R_{\text{eff}G}(a, b) \leq R_{\text{eff}H}(a, b).}$$

This is useful for proving transience. If we find a subgraph $H \subset G$ such that

$$R_{\text{eff}H}(a, \infty) < \infty,$$

then

$$R_{\text{eff}G}(a, \infty) \leq R_{\text{eff}H}(a, \infty) < \infty.$$

So G is transient.

11.2 Shorting

Shorting means identifying vertices or adding zero-resistance wires between them. Adding a zero-resistance edge gives the current more possible paths. Thus it can only decrease effective resistance.

Formally, adding an edge enlarges the class of admissible unit flows. Since Thomson's principle takes an infimum over all unit flows, enlarging the set of allowed flows can only decrease the infimum.

Therefore,

$$\boxed{\text{shorting vertices can only decrease effective resistance.}}$$

Thus if G_{short} is obtained from G by shorting some vertices, then

$$\boxed{R_{\text{eff}G_{\text{short}}}(a, b) \leq R_{\text{eff}G}(a, b).}$$

This is useful for proving recurrence. If a shorted network already has infinite resistance to infinity, then the original network must also have infinite resistance:

$$R_{\text{eff}G_{\text{short}}}(a, \infty) = \infty \implies R_{\text{eff}G}(a, \infty) = \infty.$$

12 A useful summary of the dictionary

The main correspondences are:

Random walk	Electrical network
$p(x, y) = c_{xy}/c_x$	conductance c_{xy}
$h(x) = \sum_y p(x, y)h(y)$	$\Delta h(x) = 0$
hitting probability	voltage
$\mathbb{P}_x(T_a < T_b)$	$v(x)$ with $v(a) = 1, v(b) = 0$
escape probability	conductance to infinity
$\mathbb{P}_a(T_a^+ = \infty)$	$1/(c_a R_{\text{eff}}(a, \infty))$
recurrence	$R_{\text{eff}}(a, \infty) = \infty$
transience	$R_{\text{eff}}(a, \infty) < \infty$

The core formulas are:

$$\boxed{v(x) = \mathbb{P}_x(T_a < T_b)}$$

$$\boxed{C_{\text{eff}}(a, b) = c_a \mathbb{P}_a(T_b < T_a^+)}$$

$$\boxed{\mathbb{P}_a(T_b < T_a^+) = \frac{1}{c_a R_{\text{eff}}(a, b)}}$$

$$\mathbb{P}_a(T_a^+ = \infty) = \frac{1}{c_a R_{\text{eff}}(a, \infty)}$$

$$R_{\text{eff}}(a, b) = \inf_{\theta: \text{unit flow from } a \text{ to } b} \sum_e r_e \theta(e)^2$$

cutting increases resistance, shorting decreases resistance.

13 What comes next

The machinery above reduces recurrence and transience questions to effective resistance estimates.

To prove

$\mathbb{Z}^2$ is recurrent,

we will show

$$R_{\text{eff}\mathbb{Z}^2}(0, \infty) = \infty.$$

The key tool will be shorting square shells. Shorting decreases resistance, so if the shorted network still has infinite resistance, the original network must also have infinite resistance.

To prove

$\mathbb{Z}^3$ is transient,

we will show

$$R_{\text{eff}\mathbb{Z}^3}(0, \infty) < \infty.$$

The key tool will be cutting down to a transient subnetwork, such as a nested tree $\text{NT}_{2.5849\dots}$. Cutting increases resistance, so if the subnetwork has finite resistance, then the original network has resistance at most that finite value.